control, protection, communication, and debugging fit into one design.

Takeaway. Best for applications where Hall sensors or other rotor-position sensors may not be required.

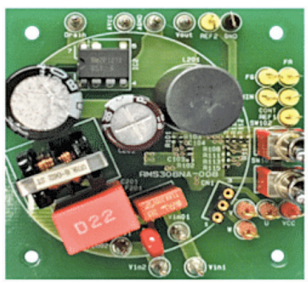
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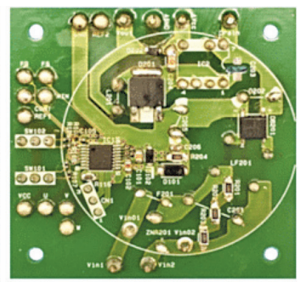
OEM Brand. NXP Semiconductors

Three-Phase sensorless BLDC Motor Drive

A 12V motor drive that plugs straight into a home AC socket

ROHM's design puts a three-phase motor drive and a 12V AC-DC converter on one board, so a 10W, 12V DC motor can run directly from a regular home socket without a separate power supply. The drive is sensorless and controls motor speed and torque. A BLDC motor has three sets of windings switched in sequence to produce rotation and, unlike brushed motors, has no brushes to wear out, which reduces maintenance.





Takeaway. Intended for low-power uses such as small ventilation fans, air purifiers, refrigerators and freezers, industrial air-cooling equipment, consumer electronics, and low-power medical equipment.

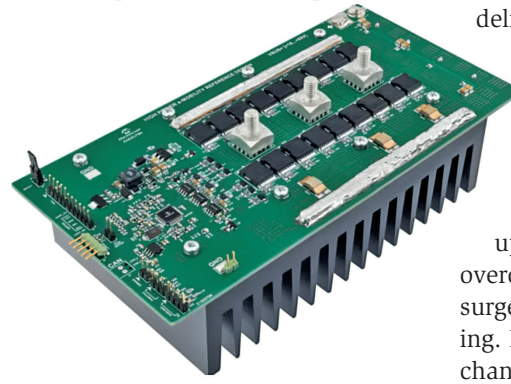
Link. *https://efy.co.in/u91hhn*

OEM Brand. ROHM

Compact 10kW BLDC Motor Controller

A standalone FOC controller with regenerative braking for systems up to 85V

Microchip's compact reference design is a standalone controller for BLDC motors and permanent magnet synchronous motors (PMSMs), with peak power up to 10kW. It uses FOC with Hall sensors and supports constant-speed, constant-torque, and constant-power



operation. Maximum Torque Per Amp (MTPA) gets the most torque from each amp in interior PMSMs (IPMSMs), and field weakening extends the constant-power range. Regenerative braking is supported, and protection covers bus overvoltage and undervoltage, motor overcurrent, inverter overtemperature, and Hall sensor errors. Onboard regulators provide 12V, 5V, and 3.3V supplies, a temperature sensor monitors the system, and CAN communication is included. An auxiliary connector provides connections for throttle input, voltage monitoring, I2C, UART, Bluetooth, and more.

Takeaway. Works for e-scooters and e-bikes, and can also run BLDC motors in high-power drones and unmanned aerial vehicles (UAVs).

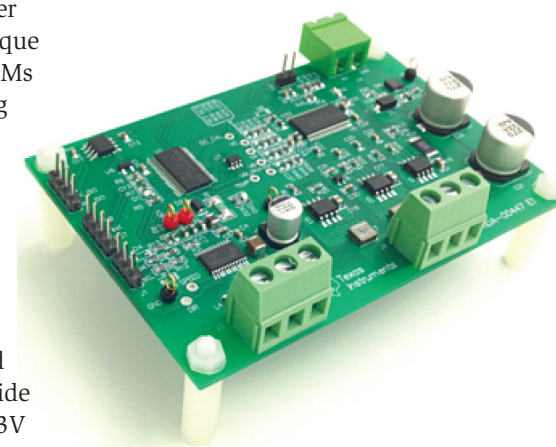
Link. *https://efy.co.in/jm8rh1*

OEM Brand. Microchip

Dual Sensorless BLDC Motor Drive

Two pump drives, 130W of continuous output, one 94 × 65mm board

Appliances often need more than one motor. Texas Instruments' (TI) design runs two low-voltage BLDC motors from a 24V DC supply: a 100W drive for a water circulation pump and a 30W drive for a drain pump. The board measures 94 × 65mm and delivers up to 130W of continuous output. The 100W drive uses a microcontroller (MCU), a three-phase gate driver, external power MOSFETs, and current sensing, with sensorless control for speeds up to 2400rpm. Protection covers overcurrent, motor lock, voltage surges, low voltage, and overheating. External MOSFETs let designers change the power section for other



motor power levels. Apart from the same protections, the 30W drive also includes a regulator.

Takeaway. Targets dishwashers and other appliances that use 24V DC pumps or fans.

Link. *https://efy.co.in/u54v9u*

OEM Brand. Texas Instruments (TI)

Together, these designs span low-power fans and appliances to e-mobility and high-power aerial systems. They cover open-loop testing (Renesas), sensorless control (NXP, ROHM, TI), AC-powered operation (ROHM), dual-motor drives (TI), and high-power FOC (Microchip). Each is a starting point for evaluation before moving to application-specific hardware. EFY

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